Fiscal Year:	FY 2006	**Task Last Updated:**	FY 10/30/2006
PI Name:	Morgan, William F. Ph.D.		
Project Title:	Mitigating High Z Radiation Induced Genomic Instability by Non-Protein Thiols		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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PI Web Page:			
City:	Baltimore	**State:**	MD
Zip Code:	21201-1509	**Congressional District:**	7
Comments:	Deceased--PI passed away November 2015		
Project Type:	Ground	**Solicitation / Funding Source:**	2004 Radiation Biology NNH04ZUU005N
Start Date:	08/12/2005	**End Date:**	07/09/2008
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: PI changed to Janet Baulch in 2008; project continues under Baulch; end date changed to 07/09/2008 (12/2010) End date change to 9/30/2009 per JSC update info (10/06)		
Key Personnel Changes/Previous PI:	add Co-I Janet E. Baulch		
COI Name (Institution):	Baulch, Janet E. (University of Maryland)		
Grant/Contract No.:	NNJ06HD31G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	That astronauts will be exposed to the space radiation environment is unavoidable, and all radiation exposure has the potential to elicit a carcinogenic response. Cancer is characterized by gross destabilization of the genome. In vivo and in vitro exposure to radiation can lead to genomic instability. This radiation induced genomic instability shares many of the characteristics associated with genetic instabilities observed in human tumors and may well be process that transforms a normal cell to a neoplastic growth following radiation exposure. We have developed a rapid and sensitive assay for measuring induced genomic instability occurring in the progeny of irradiated cells. One way to protect space travelers from the deleterious effects of radiation exposure is with the use of radioprotective compounds. In this application we will investigate the radioprotective effect of three non-protein thiol drugs already approved for clinical use. This will be achieved by investigating three specific aims: Specific aim 1 will test the hypothesis that non-protein thiols can reduce the frequency of genomic instability induced by High Z particles in human cells. These human cells have been transfected with a sensitive and reliable reporter system to analyze delayed recombination/deletion events associated with radiation exposure in the progeny of surviving cells. Specific aim 2 will test the hypothesis that some of the delayed effects of high Z radiation result from nontargeted bystander effects resulting from soluble factors secreted by irradiated cells. Specific aim 3 will investigate the molecular mechanism(s) by which the non-protein thiols exert their protective effect. Specifically we will test the hypothesis that treating cells with these drugs induces elevated intracellular levels of manganese superoxide dismutase and this reduces iron ion induced genomic instability in RKO36 cells. The goal of these studies is to determine whether drugs already approved for use in the human population can reduce genomic instability induced by radiations commonly encountered in space. If so, then these drugs can potentially be taken by astronauts to protect against deleterious effects associated with space radiations.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	These studies will test the hypothesis that non-protein thiols (NPTs) can protect astronauts from the detrimental effects of space radiation exposure and will evaluate cells surviving exposure to high Z particles to determine whether they are at increased risk for genomic instability. These studies will also evaluate the molecular mechanisms by which the NPTs exert their protective effect. The clinic can utilize a greater understanding of how these NPTs protect against ionizing radiation induced effects to improve radiotherapy treatments. Additionally, demonstrating that administration of an NPT to astronauts immediately after irradiation might reduce the detrimental effects would also have application for homeland security in the event of a radiological terrorist attack.
Task Progress:	Specific Aim 1 tests the hypothesis that non-protein thiols can reduce the frequency of genomic instability in human cells as measured by delayed recombination/deletion events using a GFP reporter assay. Specific Aim 2 tests the hypothesis that some of the effects of high Z radiation stem from non-targeted bystander effects. Specific Aim 3 evaluates the molecular mechanisms by which non-protein thiols exert their protective effects. During year 1 we have made significant progress on this project. Drs Baulch and Dziegielewski and Mr. Goetz are all now fully trained and badged for work at BNL. Additionally, Dr. Dziegielewski attended NASA's Space Radiation Summer School. We have also completed clonogenic cell survival assays to evaluate the NPT WR-1065 for cytotoxicity and low LET radio-protective effects using the RKO36 cell line. Having laid this foundation, we participated in our first beam run during September of 2006. Analyses of data and of RKO36 clones generated during this run are yielding promising preliminary results. We will further build upon these results during the Spring 2007 beam run and will apply for beam time during the Summer 2007, Fall 2007 and Spring 2008 runs. Also, as part of our efforts to characterize the molecular mechanisms of anti-oxidant drugs such as amifostine that are used to protect against and modify deleterious effects induced by ionizing radiation we have utilized the Oligo GEArray® Human NFkB Signaling Pathway Microarray by SuperArray Bioscience Corporation. It is well known that thiol containing anti-oxidant drugs in general and amifostine's active free thiol WR-1065 in particular are extremely effective in activating the transcription factor NFkB. This activation is via the reduction of cysteine residues on its p50 and p65 subunits. While NFkB is well recognized to be involved in survival pathways, it is also known to participate in inflammatory processes. For these reasons we have utilized an array system that profiles the expression of 113 genes related to NFkB mediated signal transduction. The array includes genes that encode members of the Rel, NFkB, and IkB families, NFkB-responsive genes, extracellular ligands and receptors that activate the pathway, and kinases and transcription factors that propagate the signal. This preliminary study suggests that even though NFkB is activated by WR-1065, it does not result in a robust enhancement of inflammatory genes. If the WR-1065 molecular signaling pathways had adversely affected cytokine levels and inflammatory response, it would have significantly decreased its desirability as a radioprotector.
Bibliography Type:	Description: (Last Updated: 02/02/2009)
Abstracts for Journals and Proceedings	Dziegielewski J, Baulch JE, Pyles DN, Aypar U, Murley JS, Grdina DJ, Morgan WF. "Amifostine metabolite WR-1065 protects irradiated cells from genetic instability." To be presented at the 53rd Annual Radiation Research Society Meeting, Philadelphia, PA, November 5-8, 2006. Radiat Res. 2006 Nov; supplement issue. , Nov-2006
Articles in Peer-reviewed Journals	Kataoka Y, Bindokas VP, Duggan RC, Murley JS, Grdina DJ. "Flow cytometric analysis of phosphorylated histone H2AX following exposure to ionizing radiation in human microvascular endothelial cells." J Radiat Res (Tokyo). 2006 Nov;47(3-4):245-57. Epub 2006 Sep 6. PMID: 16960336 , Nov-2006